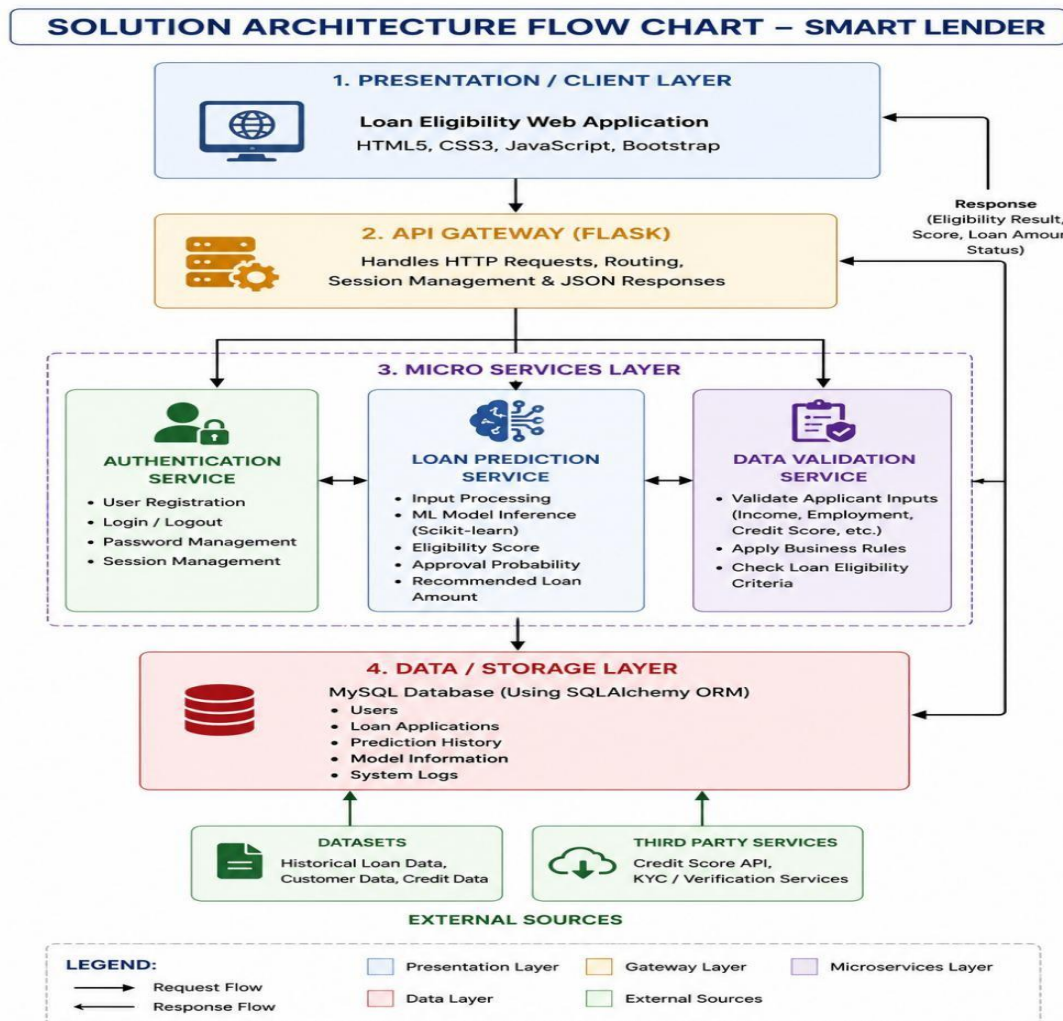


Solution Architecture

Date	30 June 2026
Team ID	
Project Name	Credit Card Approval
Maximum Marks	5 Marks

Solution Component Description Table Architecture Diagram



COMPONENT DESCRIPTION TABLE – CREDIT CARD PRO		
Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where applicants enter personal and financial details to apply for a credit card and view approval results.	HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Receives client requests, manages application routing, processes REST APIs, and controls user sessions securely.	Flask
Authentication Service	Handles secure user registration, login, multi-factor authentication (MFA), and session management for cardholders.	Python, Flask-Login
Credit Score API Service	Interacts with external credit bureaus to retrieve and parse credit reports for applicants.	Python, requests
Risk Assessment & Approval Service	Uses a trained Machine Learning model to evaluate risk factors and predict approval or denial and credit limit.	Python, Scikit-learn, TensorFlow
Identity & Fraud Verification Service	Validates applicant identities, checks against fraud databases, and enforces Anti-Money Laundering (AML) compliance.	Python, specialized third-party APIs
Decision Engine	Combines credit score, ML model output, and business rules to make final automated or manual approval decisions.	Python, Drools (or similar rule engine)
Customer Database	Stores customer data, applications, transaction history, credit decisions, and model performance metrics securely.	MySQL, SQLAlchemy ORM